DP Biology – D3.1 (HL extension)

Human reproduction in detail

- What does sexual or asexual reproduction show about change and continuity?
- How must organisms change in order to reproduce?

HL statements D3.1.13–20. Textbook pp. 709–725.
Image: a sperm meeting an egg. Public domain.

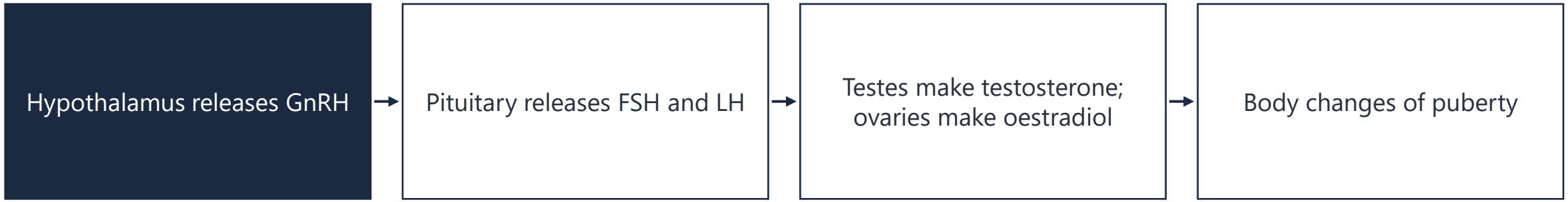


Objectives for each class

Class	Statements	By the end of the class, you can...
1	D3.1.13–14	Explain how hormones start puberty, and compare spermatogenesis and oogenesis
2	D3.1.15–17	Explain how polyspermy is prevented, how the blastocyst implants, and how pregnancy tests work
3	D3.1.18–20	Explain the role of the placenta, hormonal control of pregnancy and birth, and evaluate evidence about HRT

Hormones start puberty

D3.1.13



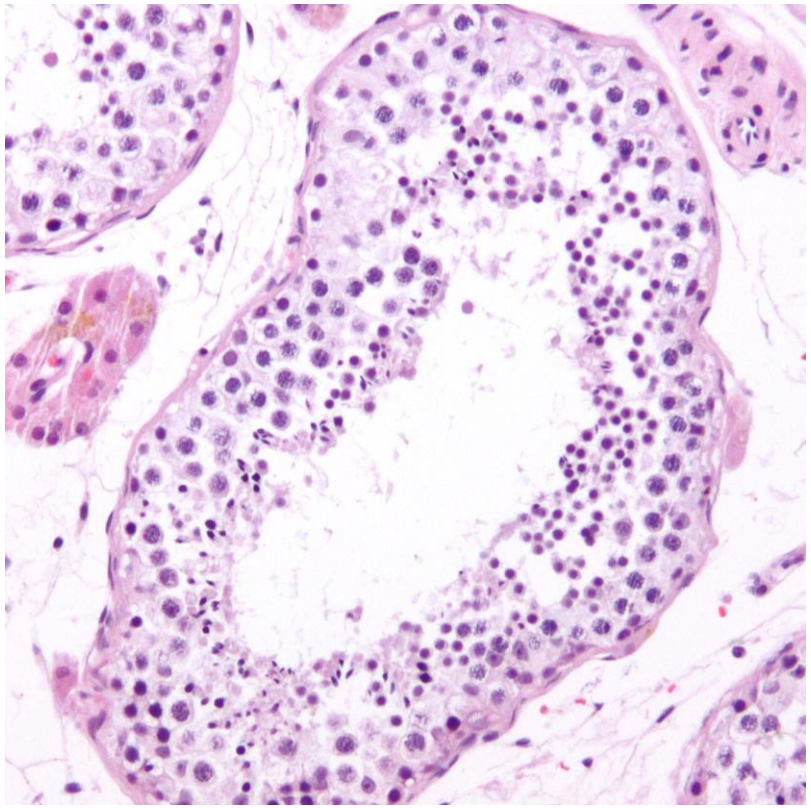
Testosterone (male)	Oestradiol (female)
sperm production starts	menstrual cycles start
growth of penis, testes	breast development
deeper voice, more muscle	wider hips
body and facial hair	body hair

Key idea

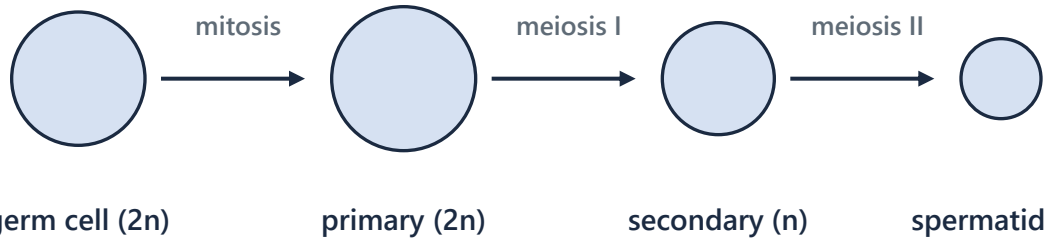
Puberty begins when the hypothalamus starts releasing GnRH in pulses. Before that, the testes and ovaries are almost inactive.

Making sperm: spermatogenesis

D3.1.14



Seminiferous tubule in section. CC BY-SA 3.0.

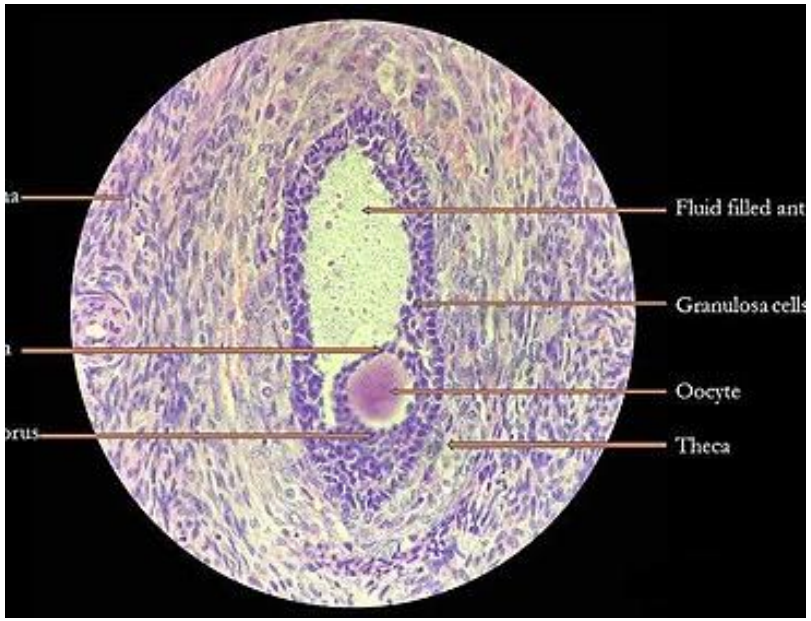


In the **seminiferous tubules**, germ cells divide by **mitosis**, grow, then go through **meiosis** to make four haploid spermatids, which develop tails.

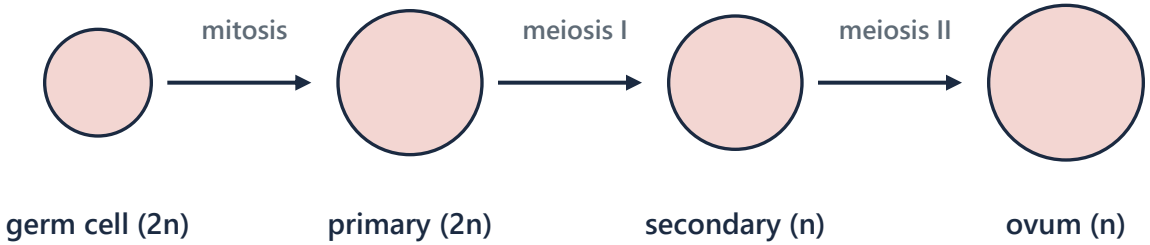
Millions are made every day from puberty onwards, starting at the edge of the tubule and moving inwards.

Making eggs: oogenesis

D3.1.14



A mature follicle with its oocyte. CC BY-SA 3.0.



Primary oocytes form **before birth** and pause in meiosis I. Each month one restarts. The divisions are **unequal**: one large cell keeps the cytoplasm, the other becomes a tiny **polar body**.

Meiosis II only finishes if a sperm enters.

Spermatogenesis and oogenesis compared

D3.1.14

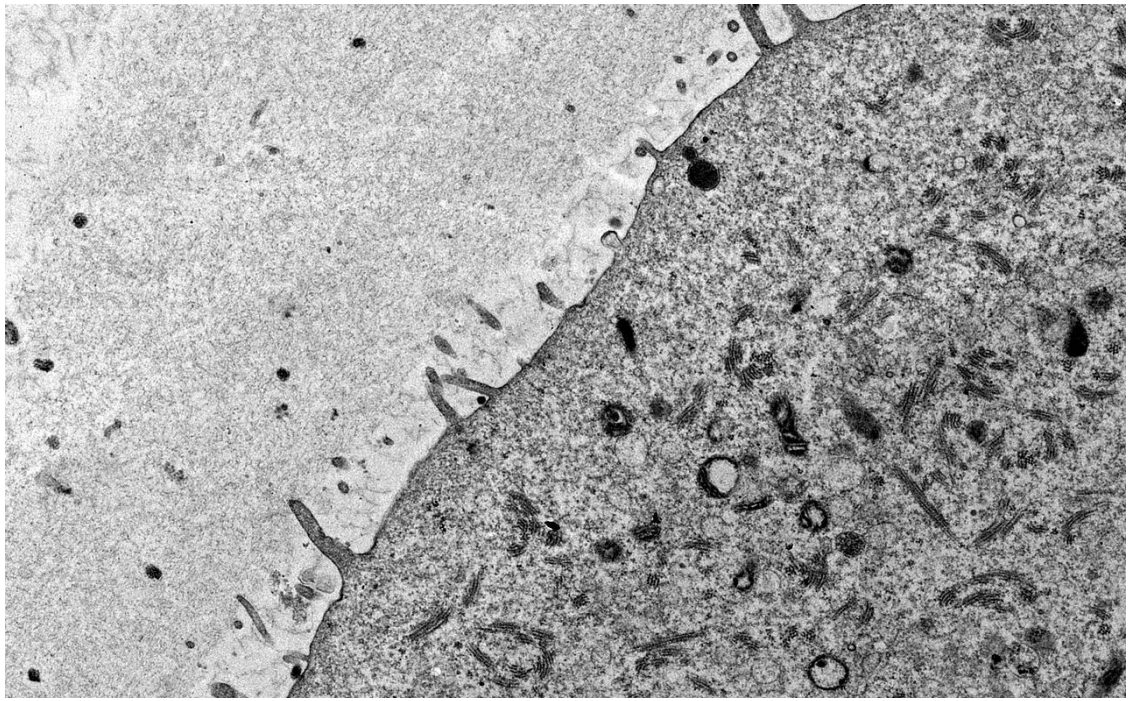
	Spermatogenesis	Oogenesis
Where	testes	ovaries
Gametes from one cell	4	1 (plus polar bodies)
Division of cytoplasm	equal	unequal
When	puberty onwards, continuous	starts before birth; one a month until menopause
Number made	millions per day	about 400 in a lifetime

Two very different gametes

D3.1.14



Human sperm. CC BY-SA 3.0.



Edge of an egg with its zona pellucida. CC0.

Sperm

Tiny, with a tail, many mitochondria and an acrosome of enzymes: built to swim and get in.

Egg

Huge, with food reserves, a protective zona pellucida and cortical granules: built to feed the embryo.

Working with data: making gametes

D3.1.14

Rounded typical values for humans.

Measure	Male	Female
Gametes per primary cell	4	1
Gametes released per month	about 3 billion	1
Years of production	about 60	about 35

1. Explain the difference in gametes per primary cell.
2. Suggest why males make so many more gametes.
3. Calculate eggs released in a woman’s life (no pregnancies).

Check-in: class 1

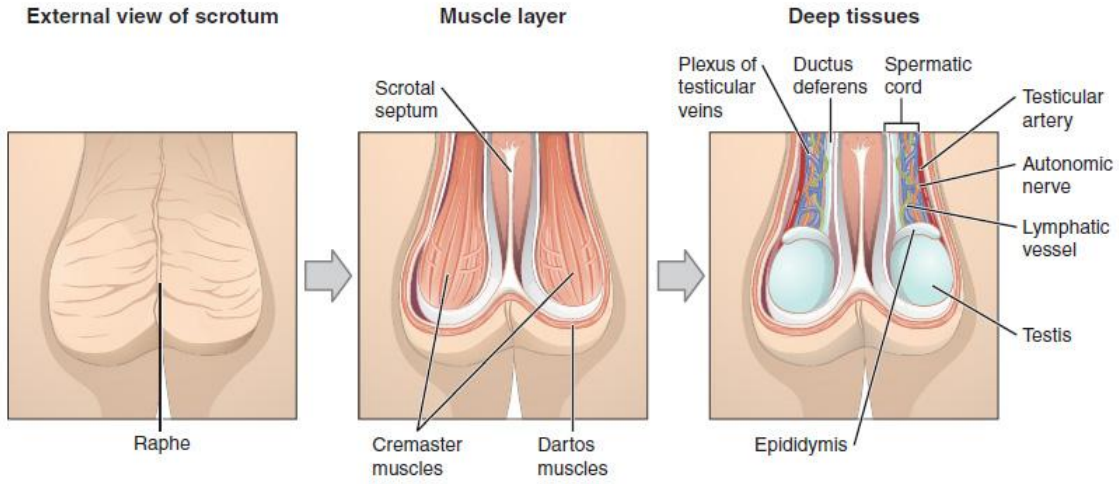
D3.1.13–14

1. Outline the hormone chain that starts puberty.
2. State two effects each of testosterone and oestradiol.
3. Name the divisions that produce sperm from a germ cell.
4. Give three differences between spermatogenesis and oogenesis.
5. Explain why oogenesis involves unequal division.

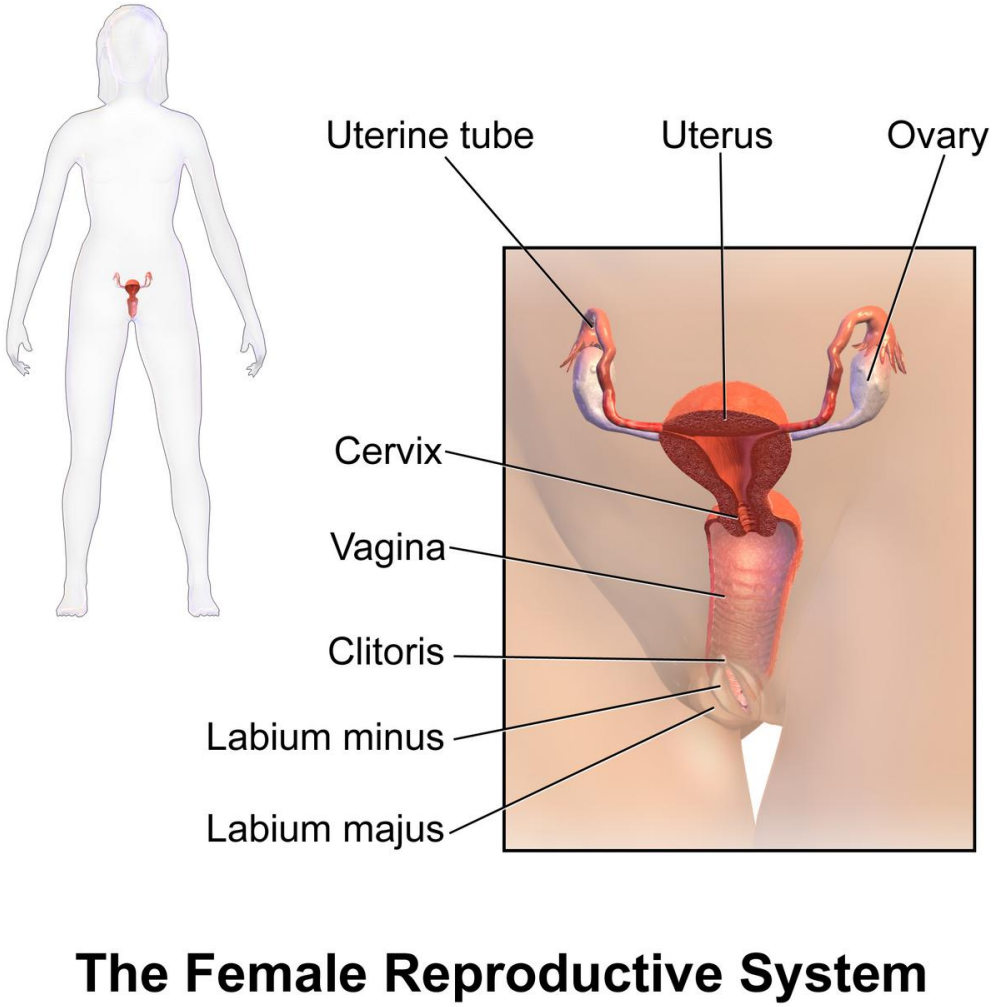
How to use this

Answer without your notes, then check against slides 3–8.

Where it happens



Male reproductive system. CC BY 3.0.

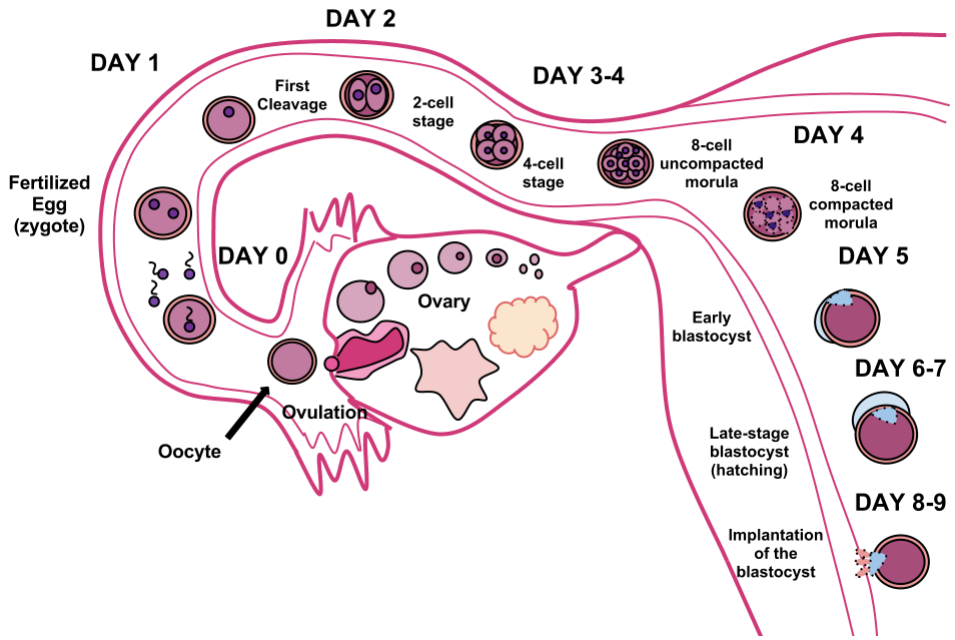


Female reproductive system. Blausen, CC BY 3.0.

Only one sperm gets in

D3.1.15

Acrosome releases enzymes → Sperm digests through the zona pellucida → Membranes fuse; sperm nucleus enters → **Cortical granules release enzymes** → Zona hardens: no more sperm



If two sperm got in, the embryo would have three sets of chromosomes and could not develop.

The **cortical reaction** changes the zona pellucida within seconds, blocking every other sperm.

Fertilization. CC BY-SA 3.0.

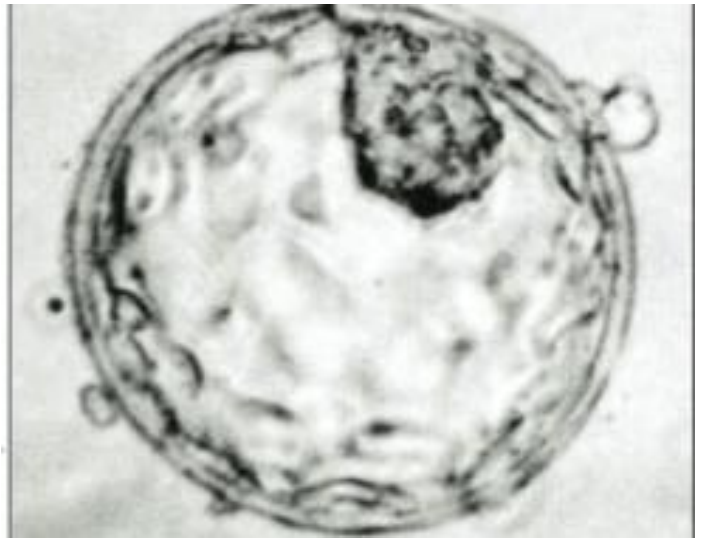
D3.1 Reproduction (HL) – Textbook pp. 716–717

From zygote to blastocyst

D3.1.16



Eight-cell embryo, day 3. Public domain.

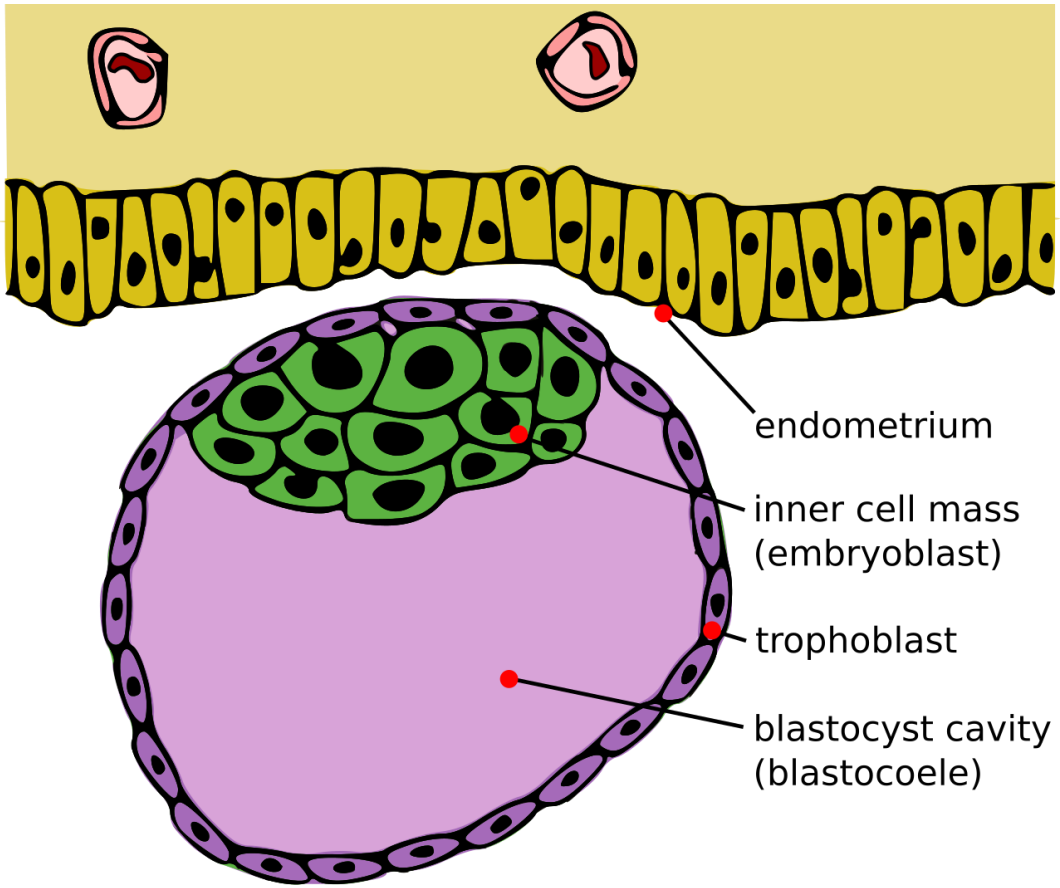


Human blastocyst, day 5. Public domain.

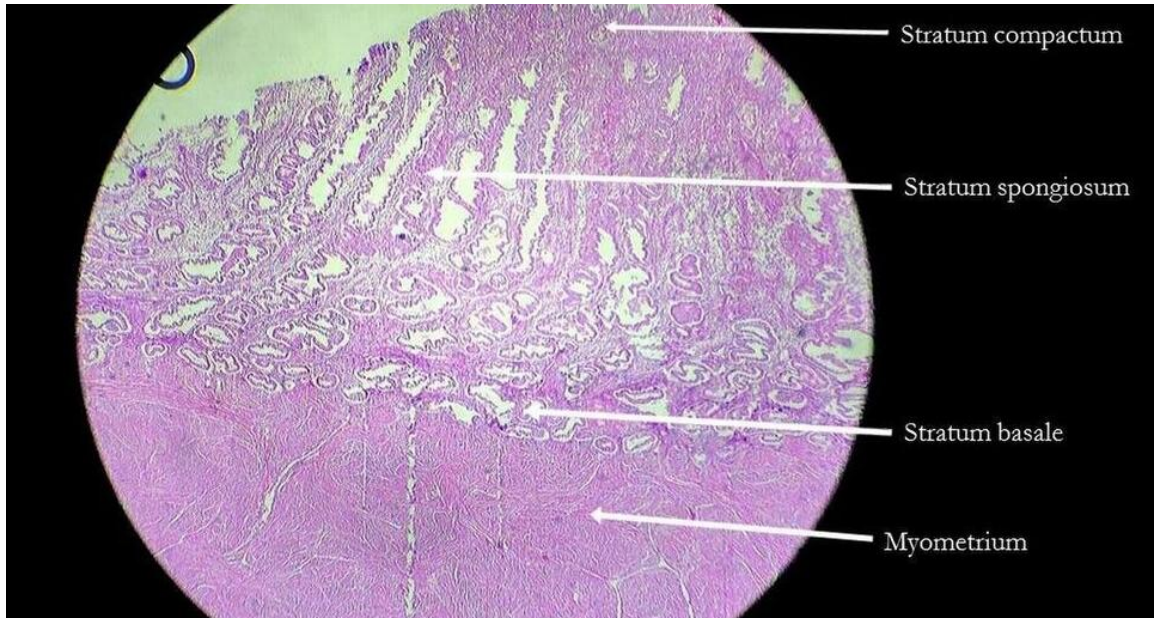
The blastocyst has an **inner cell mass** (becomes the embryo) and an outer layer that becomes part of the **placenta**.

Zygote divides by mitosis → Ball of cells moves down the oviduct → Day 5: hollow blastocyst → Day 7: implants in the endometrium

Implantation



Blastocyst implanting. CC BY-SA 3.0.

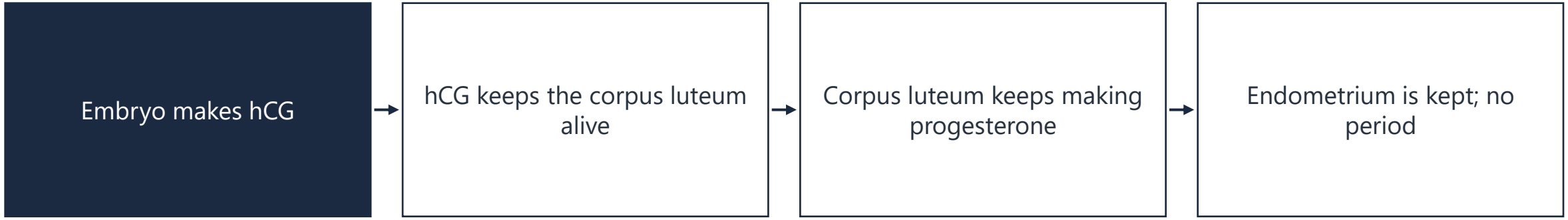


Endometrium ready for implantation. CC BY-SA 3.0.

The outer cells invade the thick, blood-rich **endometrium**. They start making the hormone **hCG**, which keeps the pregnancy going.

hCG: the first signal of pregnancy

D3.1.17, 19



Why it matters

Without hCG, the corpus luteum would break down after about two weeks, progesterone would fall and the endometrium would be shed, losing the embryo. hCG appears in the mother's blood and urine, which is what pregnancy tests detect.

How a pregnancy test works

D3.1.17



A positive test: two lines. Public domain.

Sample zone	urine soaks in
Antibodies with coloured beads	bind any hCG
Test line: fixed antibodies	catch hCG + beads: line appears
Control line	catches free beads: shows the test worked

Uses monoclonal antibodies: all identical, all binding only to hCG.

Working with data: hCG in early pregnancy

D3.1.17

Typical hCG in blood after fertilization (rounded values, IU per litre).

Days after fertilization	hCG
7	5
10	50
14	250
21	2500
28	10000

1. Describe how hCG changes.
2. A home test detects 25 IU per litre. From which day would it be positive?
3. Suggest why a test taken on day 7 might be negative even if pregnant.

Check-in: class 2

D3.1.15–17

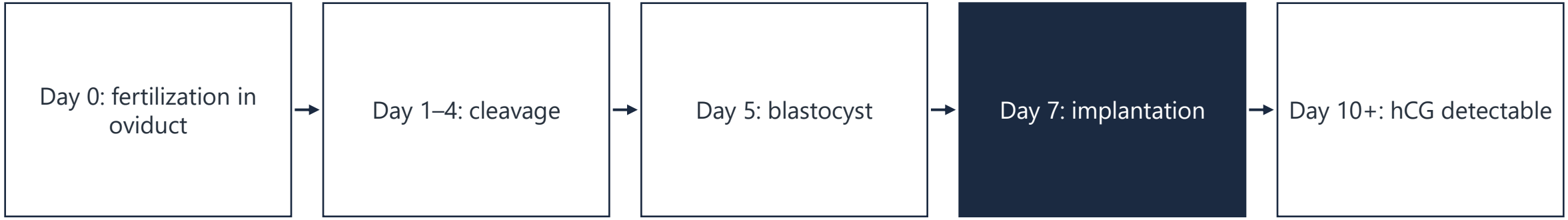
1. Explain how polyspermy is prevented.
2. Describe the blastocyst.
3. State where and when implantation happens.
4. Explain the role of hCG.
5. Explain how a pregnancy test shows a positive result.

How to use this

Answer without your notes, then check against slides 11–16.

The first two weeks

D3.1.15–17

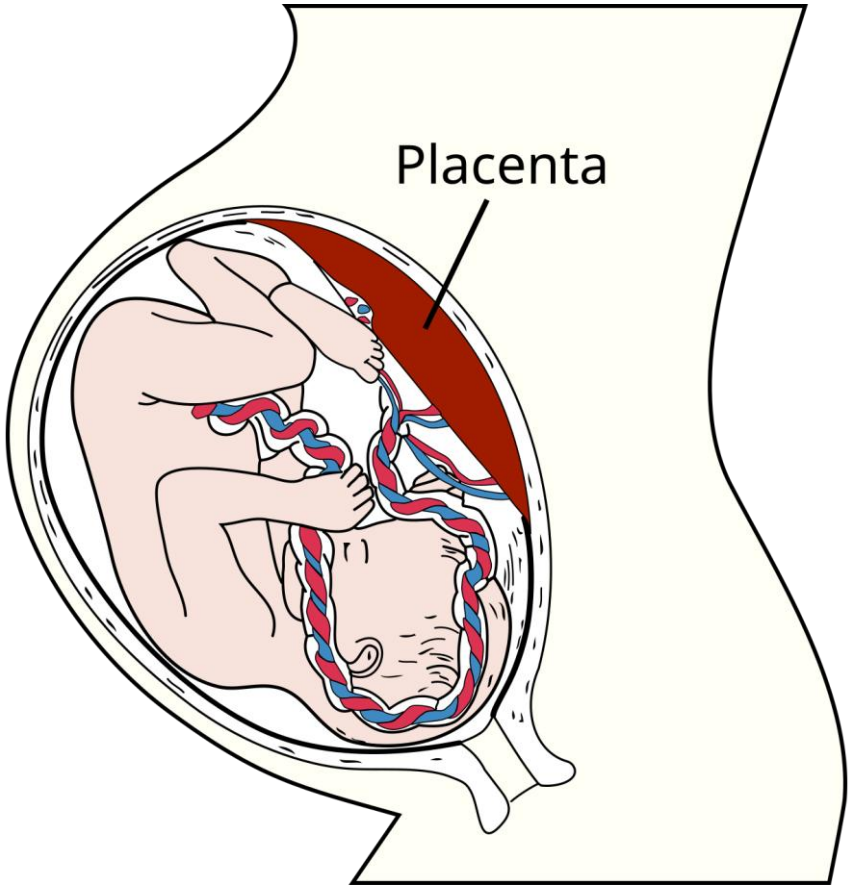


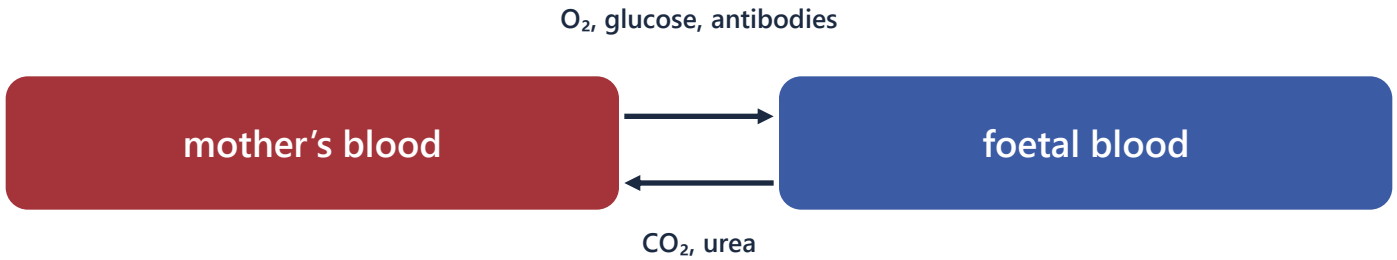
Summary

One sperm enters, the cortical reaction blocks the rest, the zygote divides into a blastocyst, the blastocyst implants and starts sending hCG, and the pregnancy can be detected.

The placenta

D3.1.18





Foetal blood runs through **villi** that sit in pools of the mother's blood. The two bloods never mix.

Thin walls and a huge surface area let substances cross by diffusion and active transport.

The placenta also makes hormones: hCG, then progesterone and oestrogen.

Placenta and umbilical cord. Public domain.

Hormones keep the pregnancy going

D3.1.19

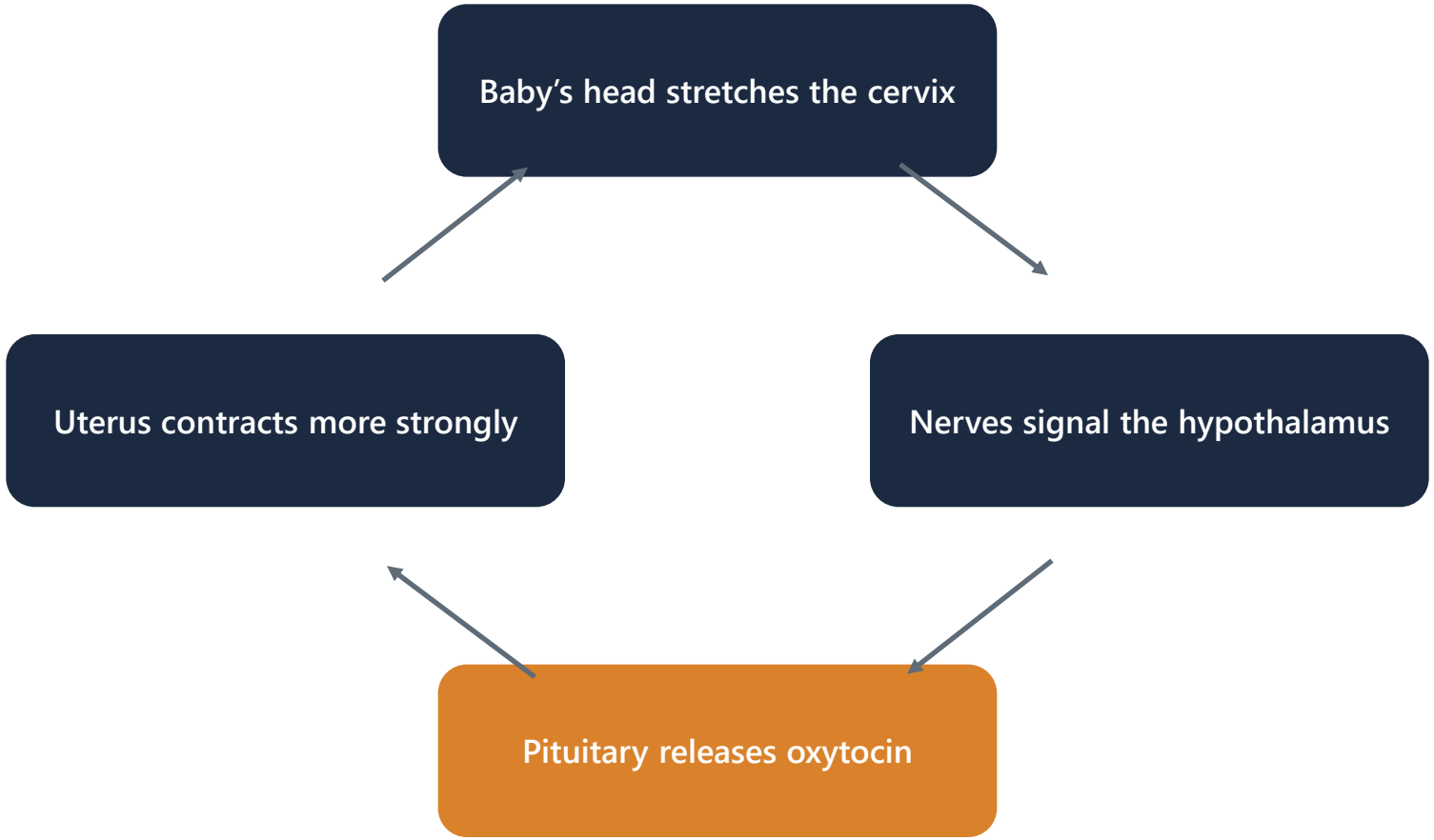
Hormone	Made by	Job
hCG	embryo, then placenta	keeps the corpus luteum alive in early pregnancy
Progesterone	corpus luteum, then placenta	keeps the endometrium; stops uterus contracting
Oestrogen	placenta	grows the uterus; prepares it for birth

Hand-over

By about 12 weeks the placenta makes enough progesterone on its own, and the corpus luteum is no longer needed.

Birth: positive feedback

D3.1.19



Each contraction leads to more oxytocin and stronger contractions, until the baby is born and the loop stops.

HRT and heart disease

D3.1.20

After menopause, oestrogen and progesterone fall. **Hormone replacement therapy (HRT)** tops them up to ease symptoms.

Early **observational** studies found HRT users had **less** coronary heart disease. Later **randomised trials** found **no protection** and a small **increase** in risk.

Why the difference?

Women who chose HRT were on average wealthier and healthier anyway (a confounding variable). Only a randomised controlled trial, which assigns treatment by chance, removes this bias. Correlation is not causation.

Working with data: two kinds of study

D3.1.20

Coronary heart disease cases per 10 000 women per year. Rounded values, based on the pattern in published studies.

Study type	HRT users	Non-users
Observational	30	45
Randomised trial	37	30

1. Compare the two studies.
2. Explain why the observational study was misleading.
3. Explain why the trial is more reliable.

Practice: apply the ideas

D3.1.18–20

Smoking in pregnancy

Suggest why babies of smokers are often smaller.

Labour too slow

Explain why doctors may give oxytocin.

Corpus luteum removed at week 6

Predict what would happen to the pregnancy.

How to use this

Answer each in two or three sentences.

Common mistakes to avoid

Not this	But this
Oogenesis gives four eggs.	One egg plus polar bodies.
Mother’s and baby’s blood mix in the placenta.	They stay separate; substances cross thin walls.
A pregnancy test detects progesterone.	It detects hCG.
Birth is controlled by negative feedback.	Oxytocin release is positive feedback.

Check that you can...

1. **Outline** the hormones that start puberty
2. **Compare** spermatogenesis and oogenesis
3. **Explain** how polyspermy is prevented
4. **Describe** blastocyst formation and implantation
5. **Explain** how a pregnancy test works
6. **Explain** the role of the placenta and hormones in pregnancy and birth
7. **Evaluate** evidence about HRT and heart disease

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. State two differences between spermatogenesis and oogenesis.
2. Explain how a pregnancy test detects hCG.
3. Explain why childbirth is an example of positive feedback.

Image credits

Sperm-egg.jpg – Unknown author – Public domain
Seminiferous tubule and sperm low mag.jpg – Nephron – CC BY-SA 3.0
Graafian follicle.jpg – Dr. Roshan Nasimudeen – CC BY-SA 3.0
Spermatozoa.jpg – Tomas Drab – CC BY-SA 3.0
Zona pellucida.jpg – Prue Talbot – CC0
Figure 28 01 02.JPG – OpenStax College – CC BY 3.0
Blausen 0399 FemaleReproSystem 01.png – BruceBlaus – CC BY 3.0
Human Fertilization.png – Ttrue12 – CC BY-SA 3.0
Embryo, 8 cells.jpg – ekem, Courtesy: RWJMS IVF Program – Public domain
Human blastocyst.jpg – Mr. J. Conaghan – Public domain
Blastocyst English.svg – Seans Potato Business (derivative of the source cited above) – CC BY-SA 3.0
Secretory endometrium.jpg – Dr. Roshan Nasimudeen – CC BY-SA 3.0
Pregnancy test.jpg – Klaus Hoffmeier – Public domain
Placenta.svg – Gray38.png: User Magnus Manske on en.wikipedia derivative wo – Public domain

All from Wikimedia Commons (commons.wikimedia.org, search the file name). Some images are cropped. CC licences: creativecommons.org/licenses